VLSI PROJECT:

DIFFERENTIAL AMPLIFIER WITH ACTIVE LOAD USING CROSS-COUPLED BULK IN GPDK45

IMPLEMENTED IN:

CADENCE VIRTUOSO

NAME:

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ROLL NO: S20240020317

TARGET:

Differential gain>40dB

Bandwidth>280Khz

CMRR>40 with minimum power

INTRODUCTION

A CMOS differential amplifier is an analog circuit used to amplify the difference between two input signals while rejecting signals that are common to both inputs. Cross-coupled bulk connections enhance the differential amplifier's common-mode rejection while enabling low-power operation. The two important operating modes are **differential mode** and **common-mode** operation. In differential mode, the two input signals have equal magnitude and opposite phase, whereas in common-mode operation, the same signal is applied to both inputs.

The differential output is given by

$$V_{od} = V_{out1} - V_{out2}$$

and the differential input is

$$V_{id} = V_{in1} - V_{in2}.$$

Therefore, the differential gain

$$\boxed{A_d = \frac{V_{out1} - V_{out2}}{V_{in1} - V_{in2}}}$$

For common-mode operation, the same signal is applied to both inputs. The common-mode input and output are defined as

$$V_{icm} = \frac{V_{in1} + V_{in2}}{2}$$

$$V_{ocm} = \frac{V_{out1} + V_{out2}}{2}$$

$A_{cm} = V_{ocm} / V_{icm}$

$CMRR = A_d / A_{cm}$

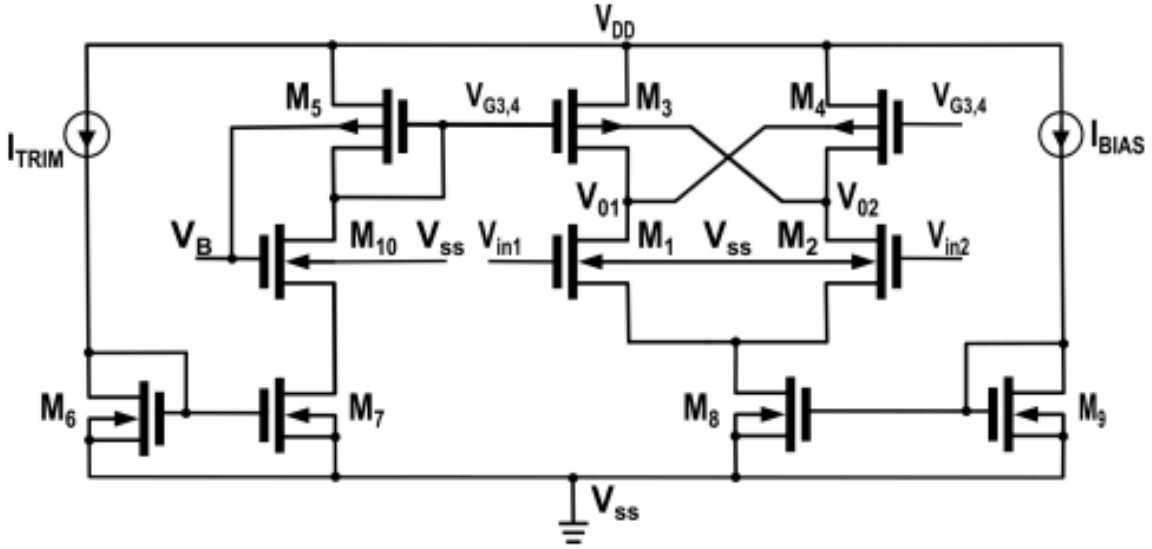
The transconductance is:

$$g_m = \sqrt{2\mu C_{ox} \frac{W}{L} I_D}$$

For a fixed channel length and drain current:

$$g_m \propto \sqrt{W}$$

SCHEMATIC DIAGRAM

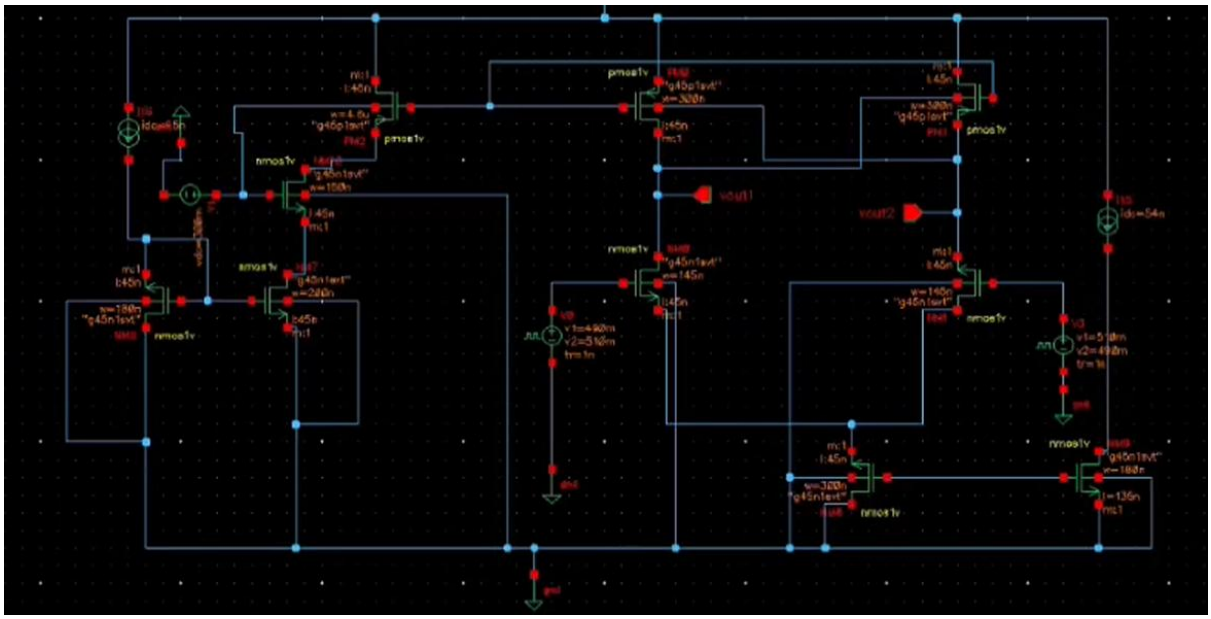


A Fully Differential Amplifier with active load using cross-coupled bulk to enhance the gain and CMRR Kashif Nisar1 , Andrea Ballo2 , Member, IEEE, Alfio Dario Grasso3 , Senior Member, IEEE Dipartimento di Ingegneria Elettrica, Elettronica e Informatica (DIEEI), University of Catania 1kashif.nisar@phd.unict.it, 2 andrea.ballo@unict.it, 3 alfiodario.grasso@unict.it

In this design, the channel length was kept constant at 45 nm, while the transistor widths were optimized through simulation. Different transistors were resized based on their function within the amplifier:

Transistor Group	***Primary Effect***
M1, M2	*Increase transconductance and differential gain*
M3, M4	*Improve common-mode rejection and CMRR*
M5	*Maintain biasing and common-mode operating point*
M8	*Increase output resistance and reduce common-mode gain*

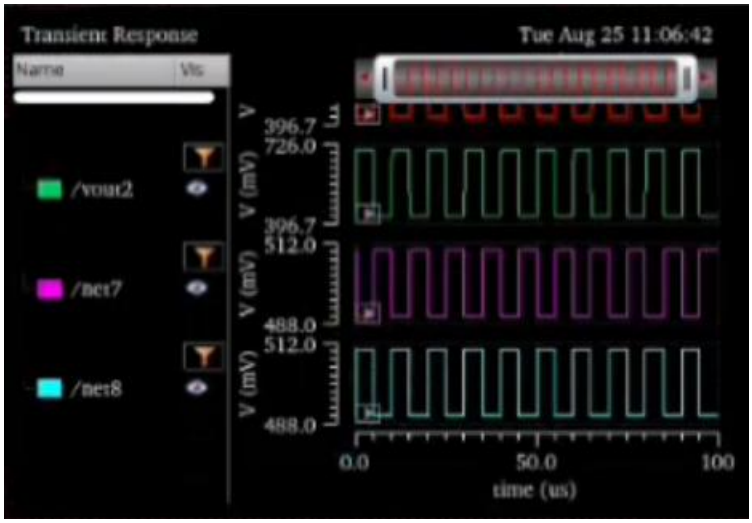
SCHEMATIC



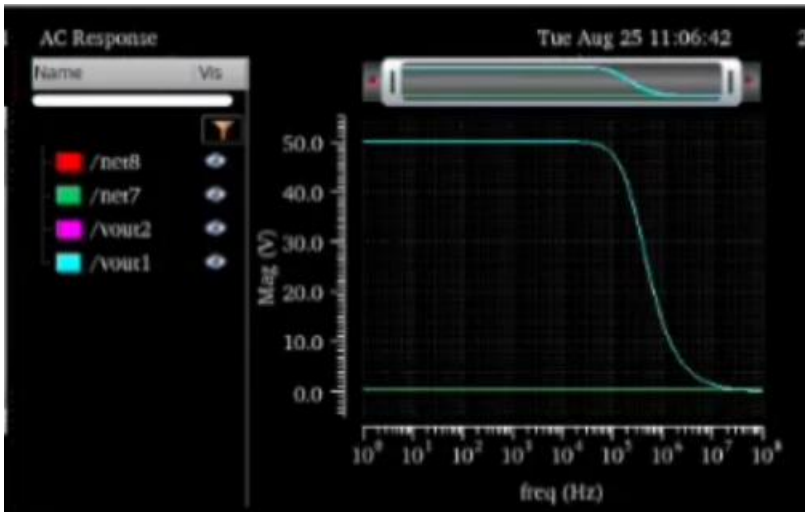
Transistor	***Width*** W	***Length*** L		
M1	*145 nm*	*45 nm*	*VDD*	*1V*
M2	*145 nm*	*45 nm*	*ITRIM*	*54nA*
M3	*300 nm*	*45 nm*	*IBIAS*	*95nA*
M4	*300 nm*	*45 nm*		
M5	*4.6 µm*	*45 nm*		
M6	*180 nm*	*45 nm*		
M7	*200 nm*	*45 nm*		
M8	*300 nm*	*45 nm*		
M9	*180 nm*	*135 nm*		
M10	*150 nm*	*45 nm*		

ANALYSIS:

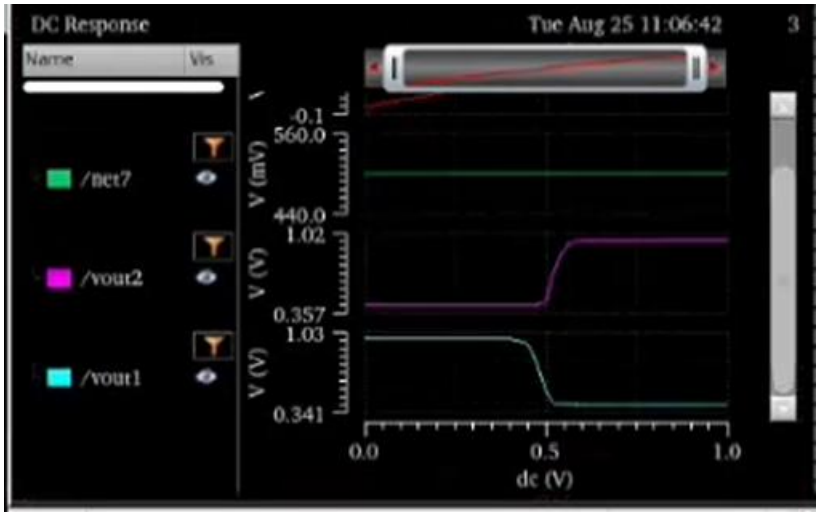
TRANSIENT RESPONSE:



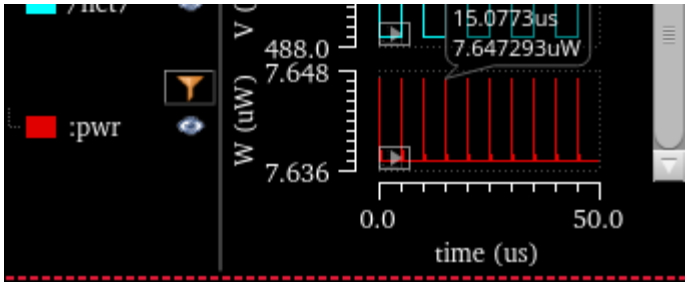
AC RESPONSE:



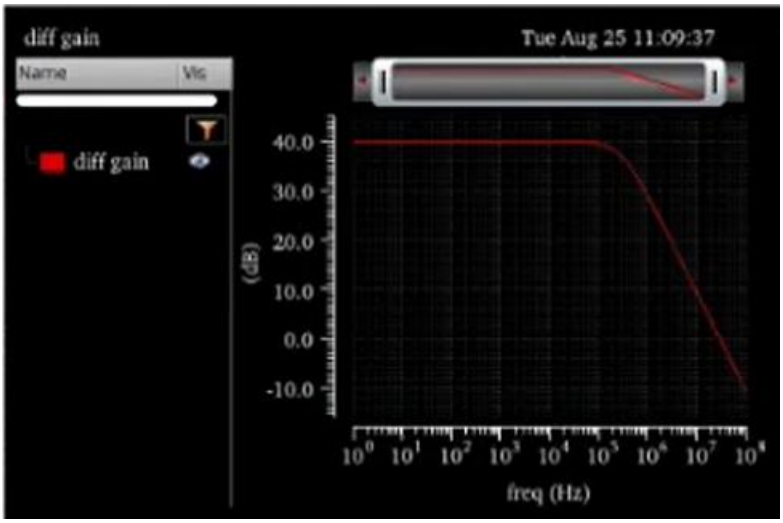
DC RESPONSE:



POWER:7.64uW

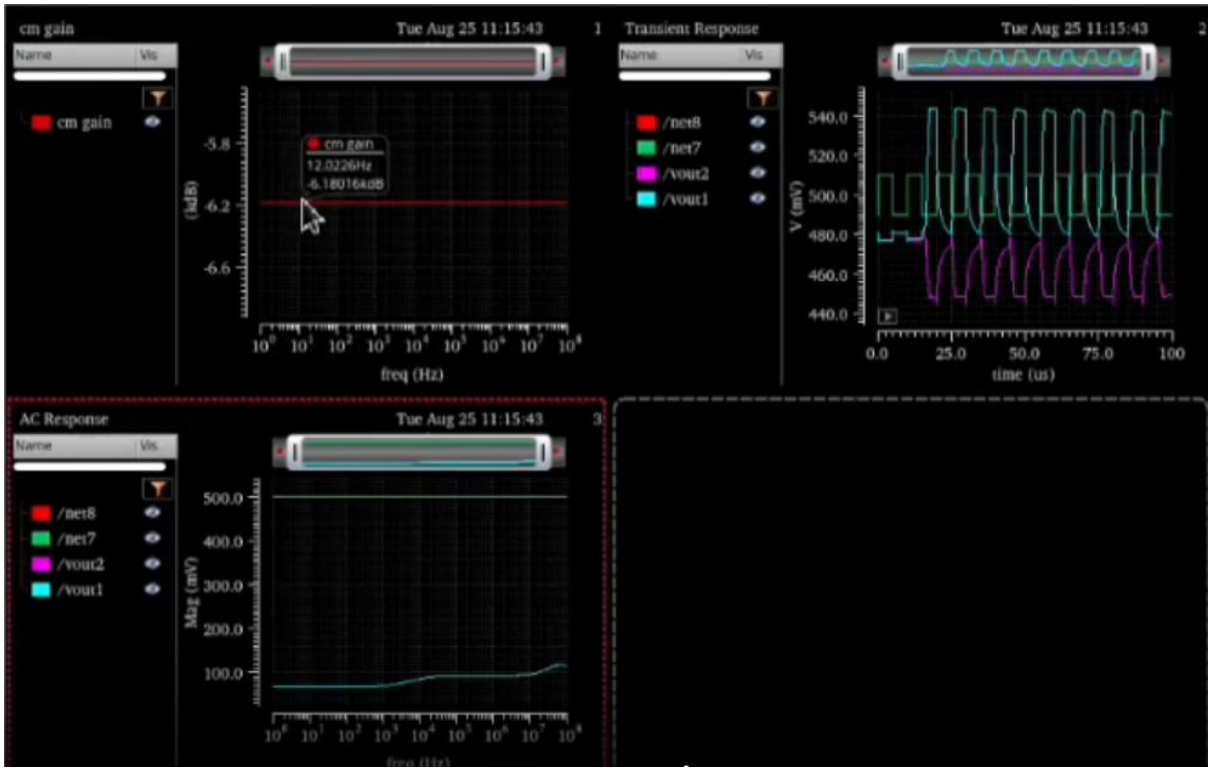


DIFFERENTIAL GAIN: 40.03dB

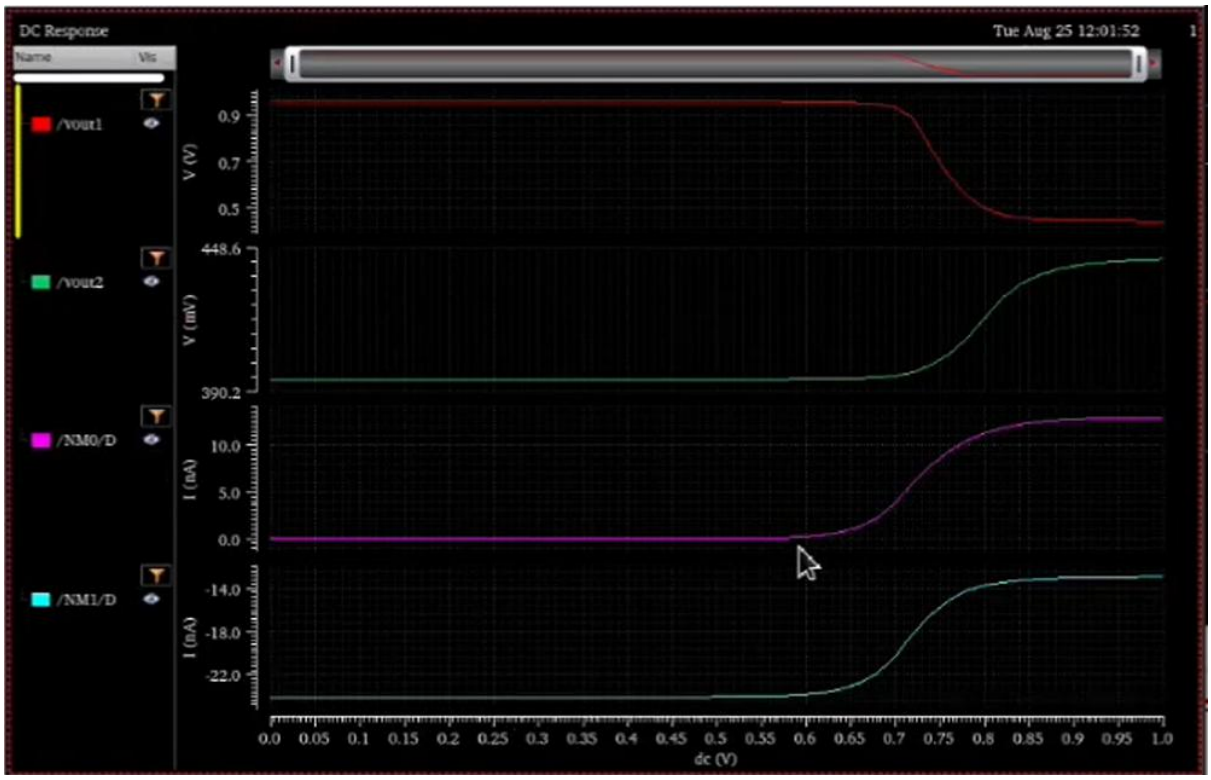


BANDWIDTH: 290KHZ

COMMON-MODE GAIN: -6.21dB

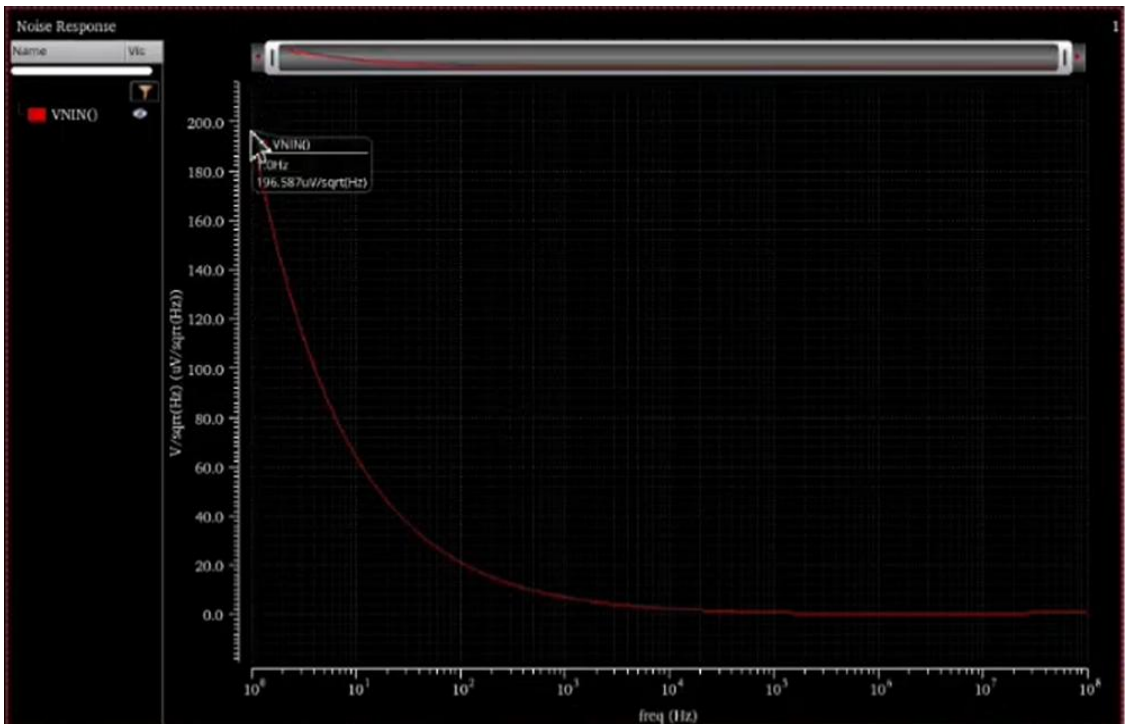


DC SWEEP (Variable = vcm):

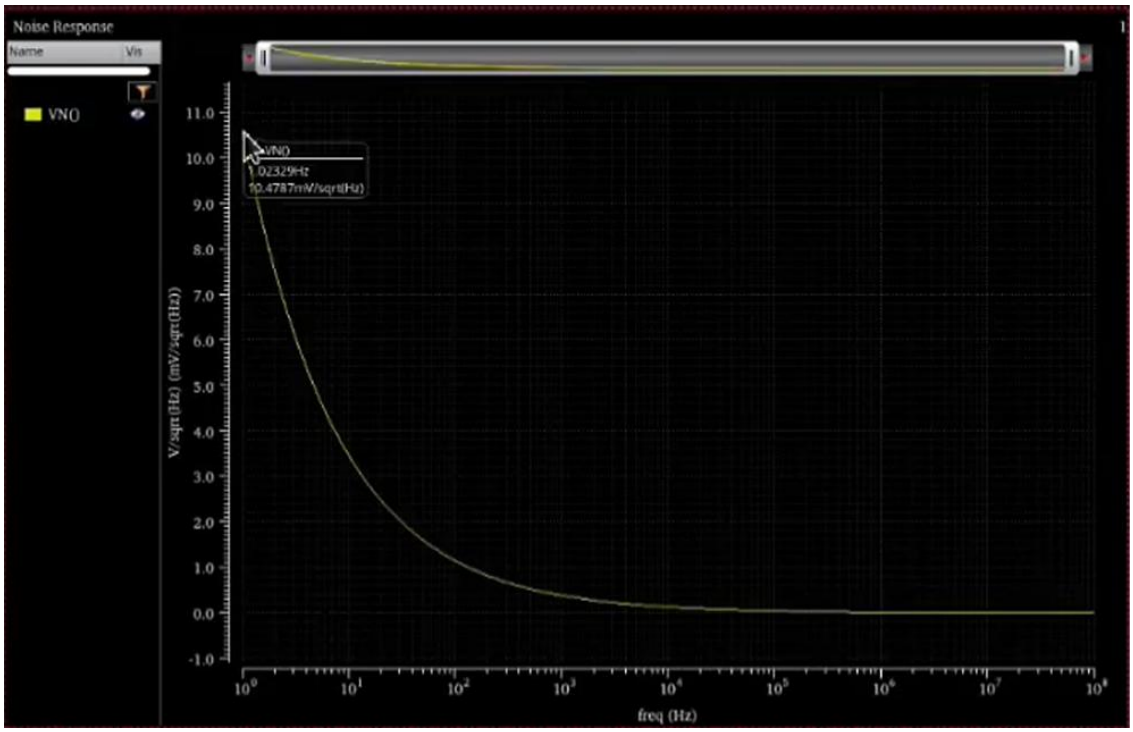


NOISE:

INPUT EQUIVALENT NOISE GRAPH:



OUTPUT EQUIVALENT NOISE GRAPH:



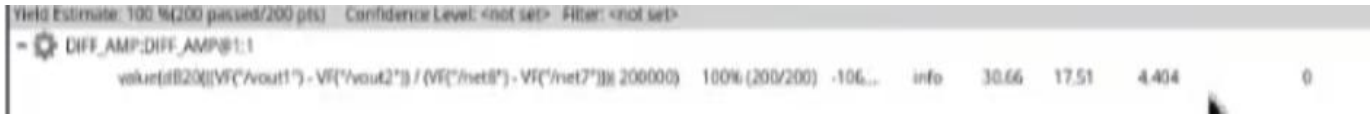
CMRR :

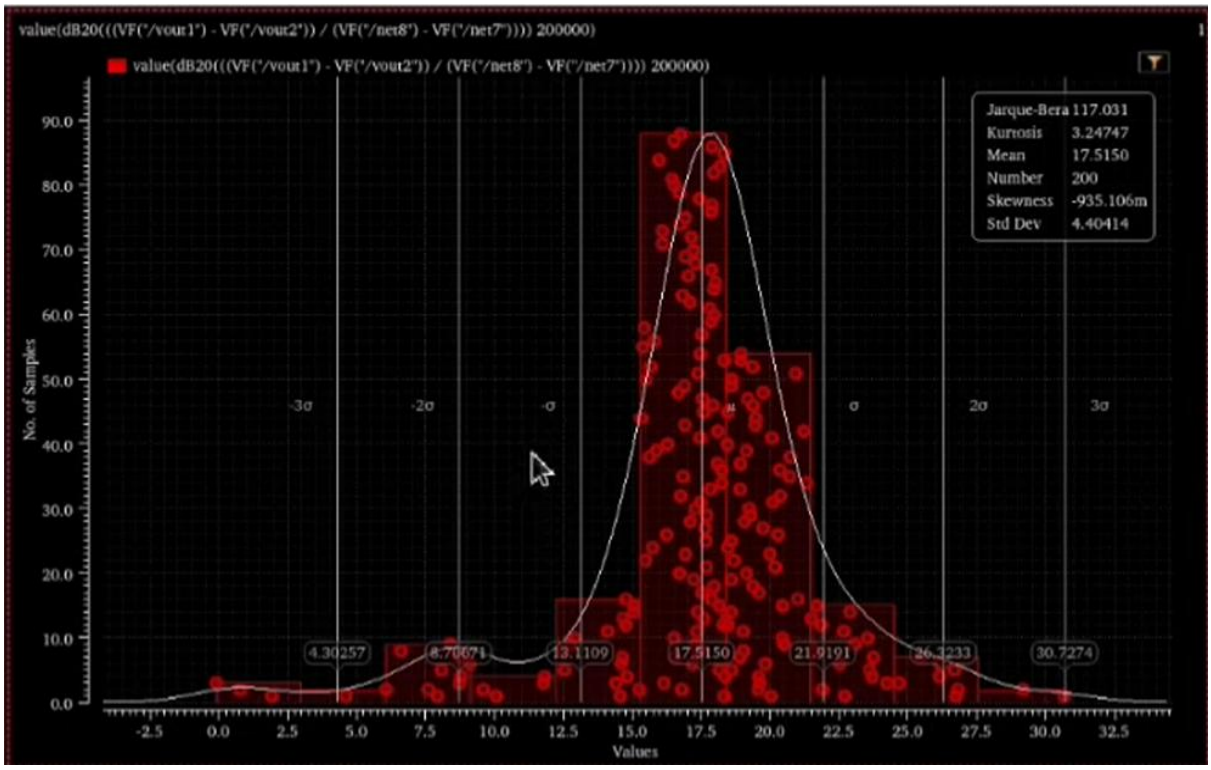
Differential gain – common mode gain (in dB)

40.03 – (-6.21) = 46.24dB

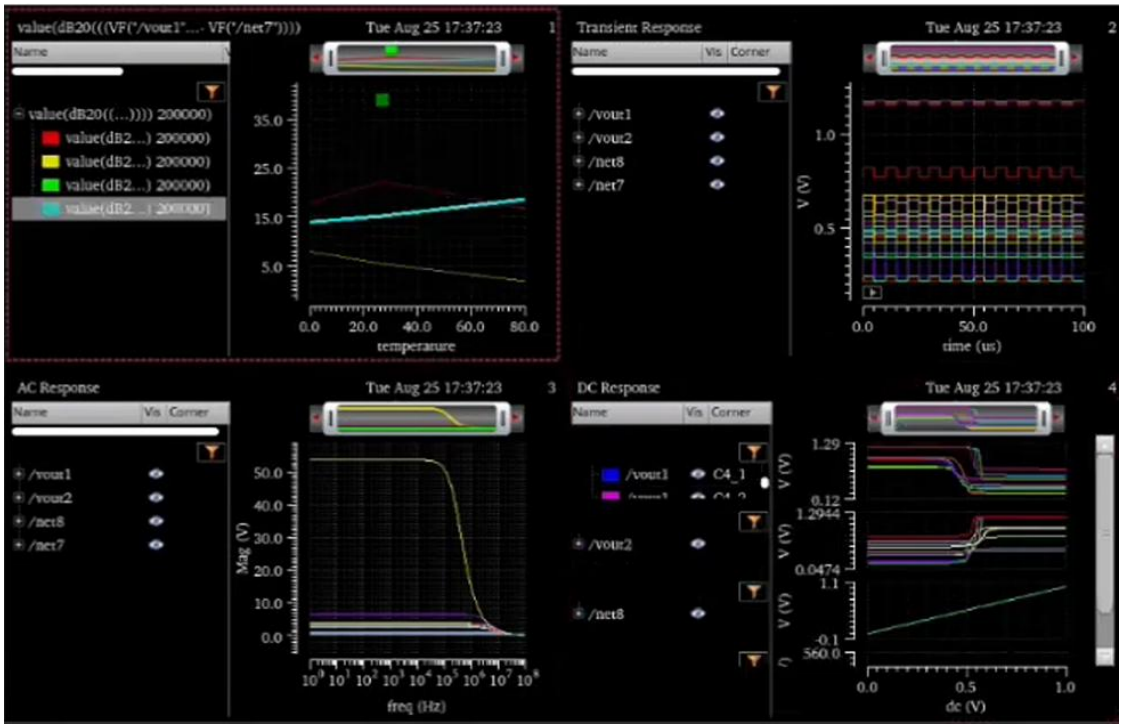
MONTE CARLO (Differential gain at 2MHZ):

A Monte Carlo analysis with 200 samples was performed to evaluate the effect of random process variations and device mismatch on the differential gain. The distribution provides the mean value and standard deviation of the gain, allowing the robustness of the proposed design to be evaluated.





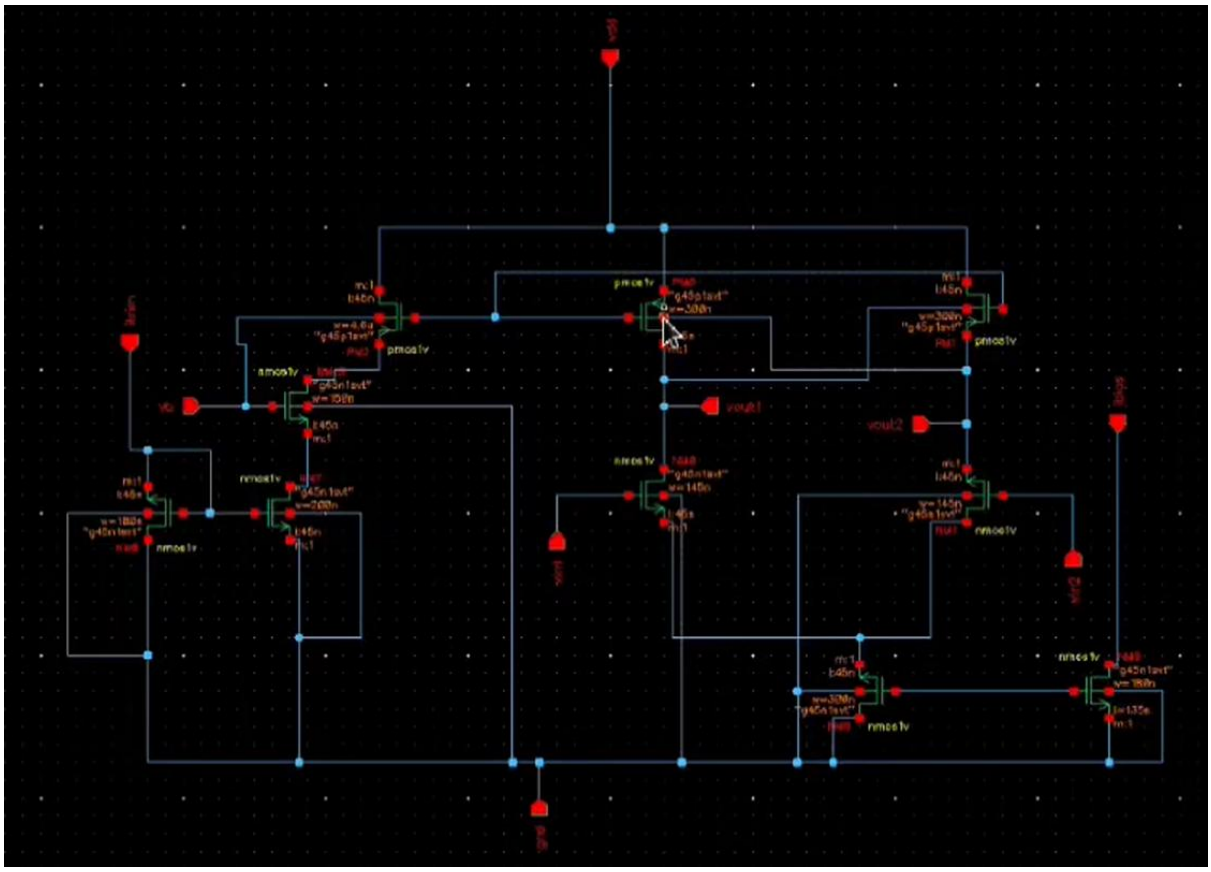
PVT :



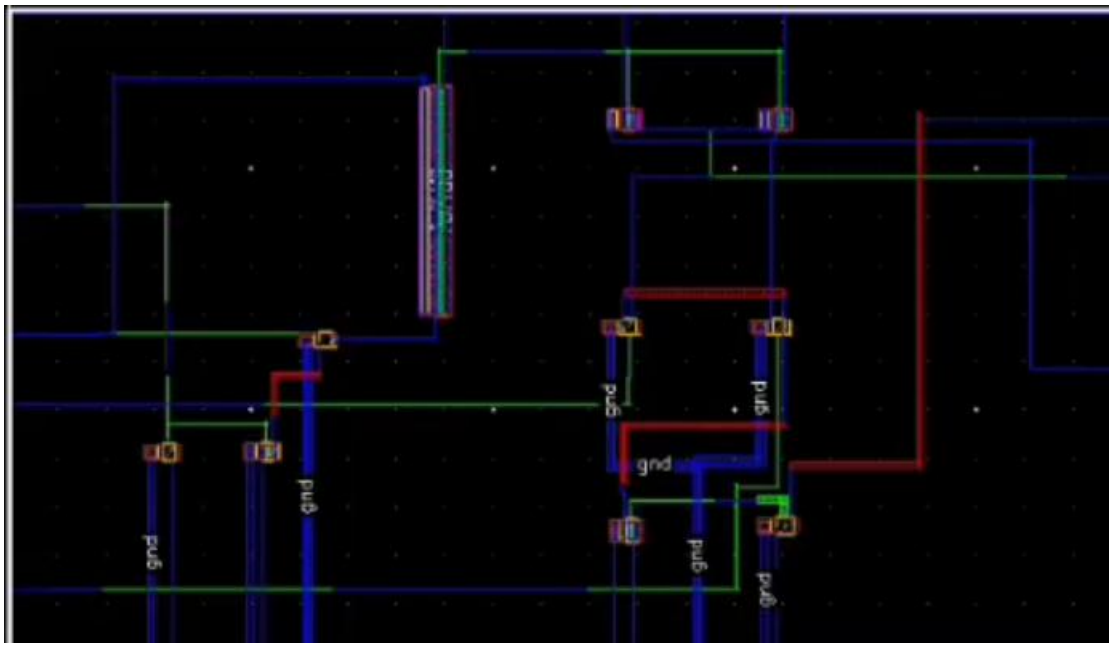
Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C3_0	C3_1	C3_2	C4_0	C4_1	C4_2	C5_0	C5_1	C5_2
DIFF_AMP:DIFF_AMP@1:1	/vout1															
DIFF_AMP:DIFF_AMP@1:1	/vout2															
DIFF_AMP:DIFF_AMP@1:1	/net8															
DIFF_AMP:DIFF_AMP@1:1	/net7															
DIFF_AMP:DIFF_AMP@1:1	value(dB20(VF(...	39.02				1.821	39.02	17.96	22.41	16.72	7.949	5.384	1.821	13.98	15.22	18.65

PVT analysis was performed to evaluate the circuit performance under variations in process corner, supply voltage and temperature. The analysis verifies the robustness of the amplifier under different operating conditions.

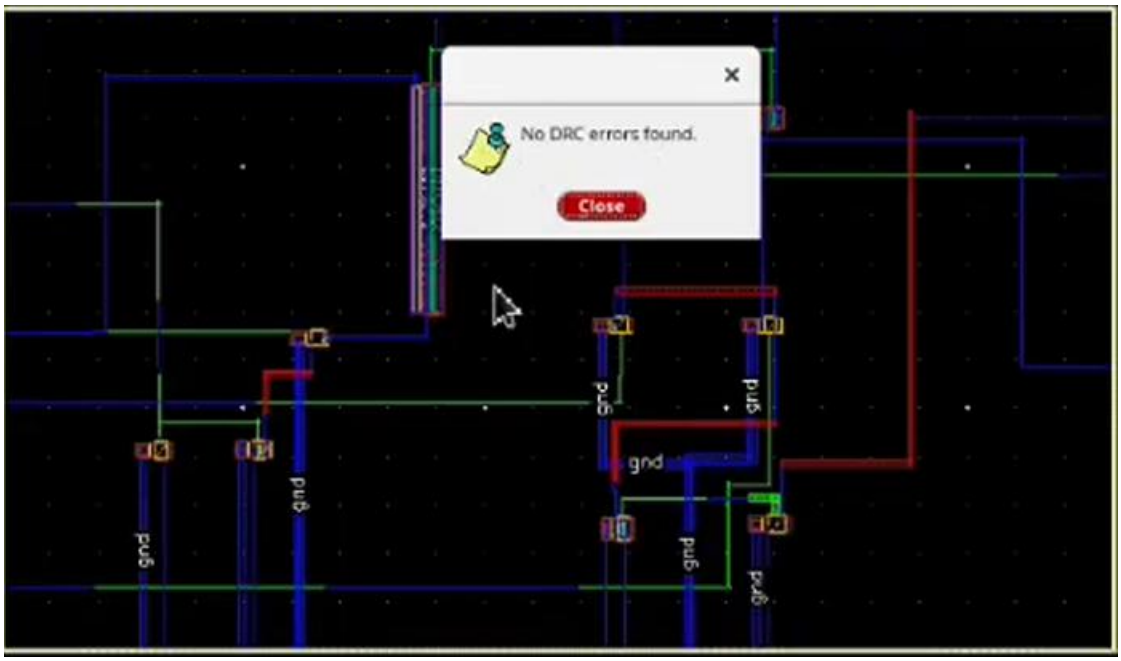
PIN DIAGRAM:



LAYOUT:



DRC CHECK:

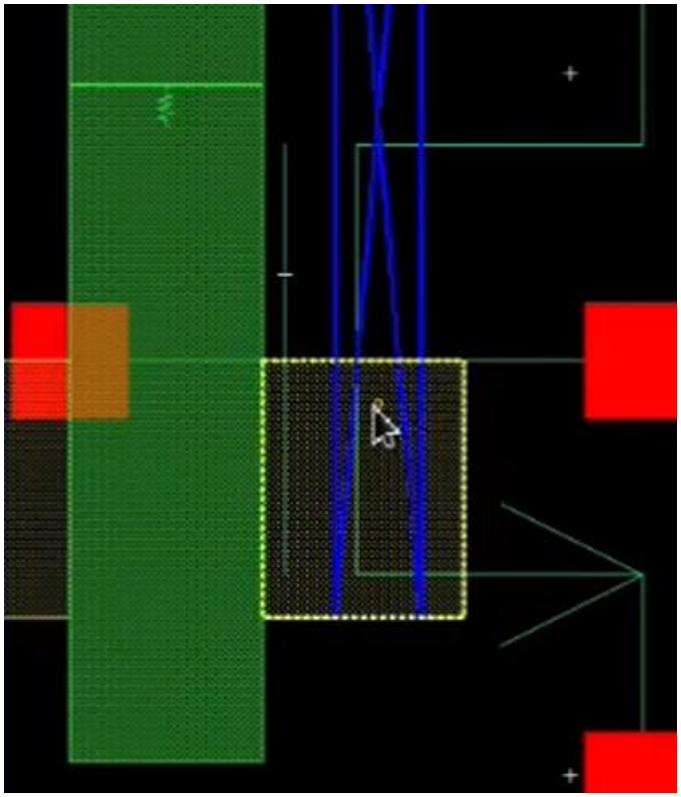


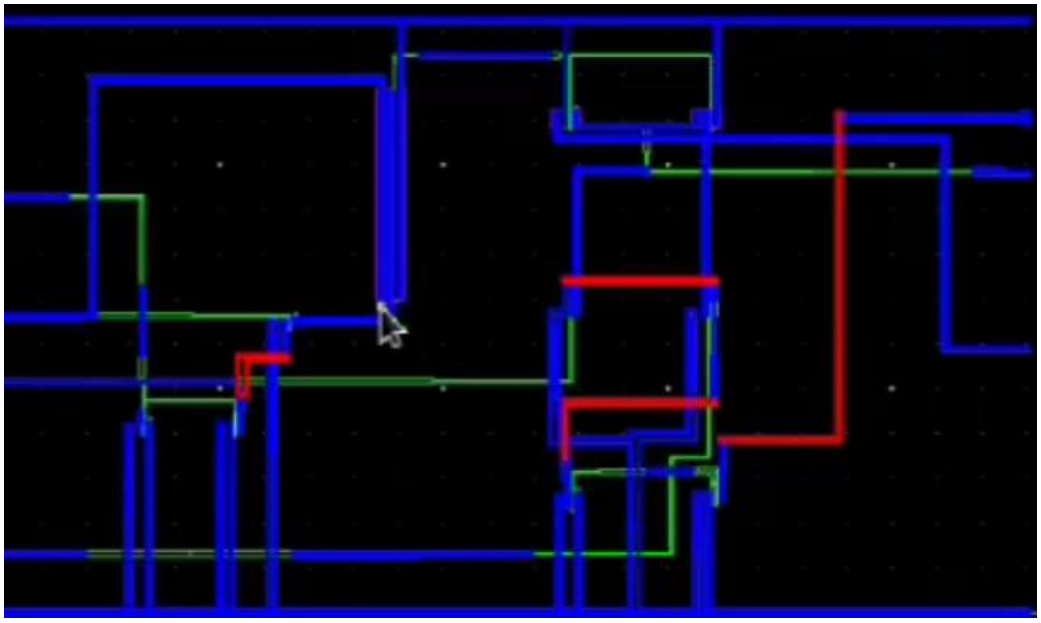
LVS CHECK:



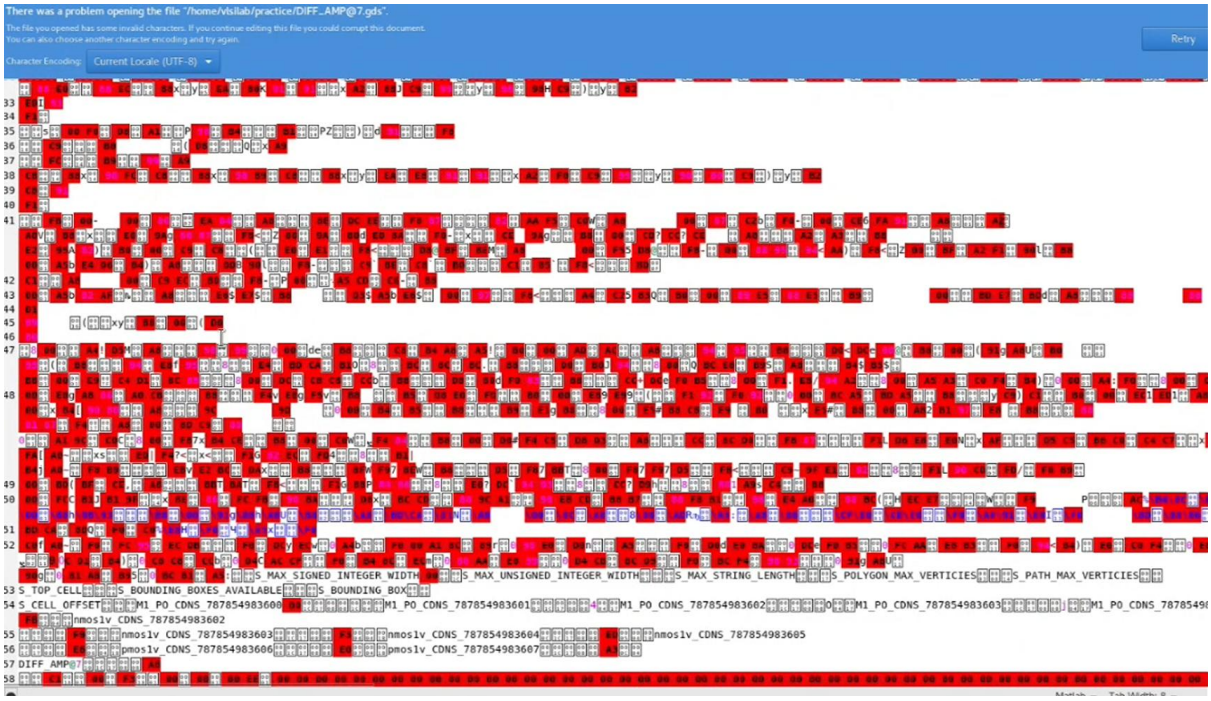


RC EXTRACTED LAYOUT:

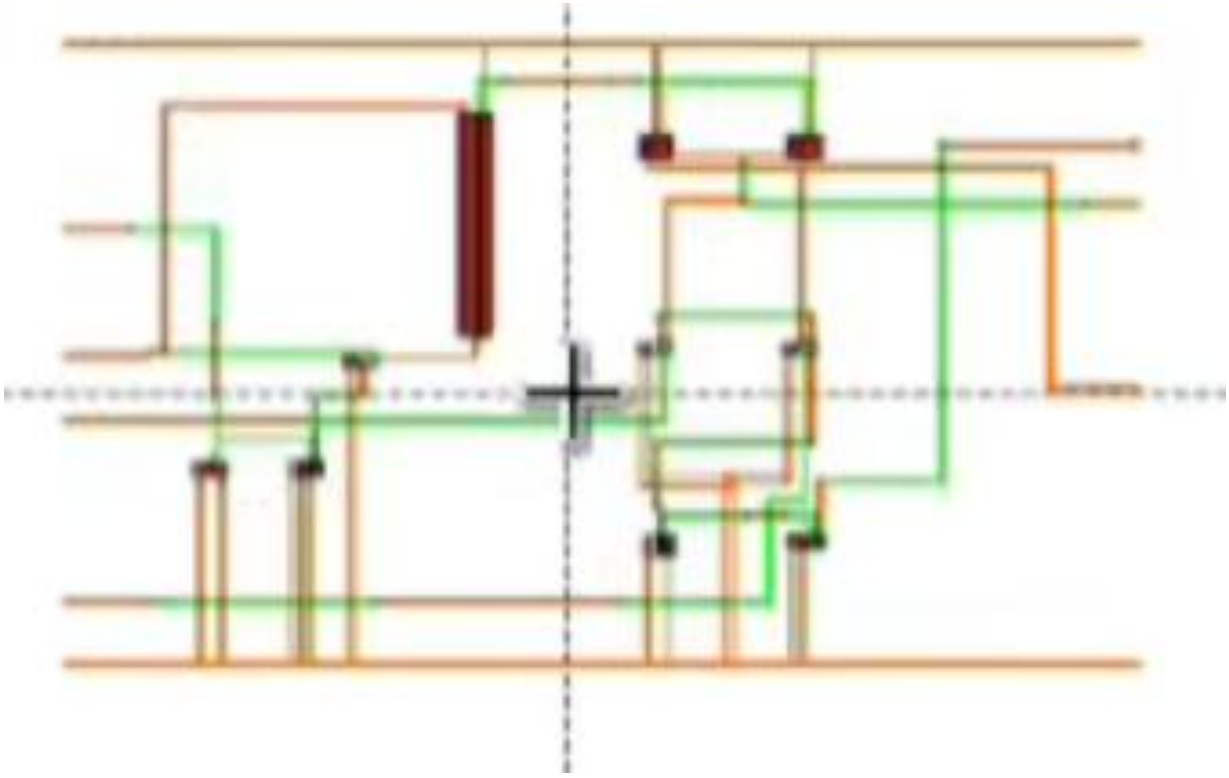




GDS FILE:



GDS LAYOUT:



REFERENCES:

1.Cadence GPDK45 Generic PDK Design Kit Documentation, Cadence Design Systems.

2. A Fully Differential Amplifier with active load using cross-coupled bulk to enhance the gain and CMRR Kashif Nisar1 , Andrea Ballo2 , Member, IEEE, Alfio Dario Grasso3 , Senior Member, IEEE Dipartimento di Ingegneria Elettrica, Elettronica eInformatica (DIEEI), University of Catania 1kashif.nisar@phd.unict.it, 2 andrea.ballo@unict.it, 3 alfiodario.grasso@unict.it